Gabriel Ona

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Master of Applied Science, University of Toronto
Edward S. Rogers Sr. Electrical and Computer Engineering

Portfolio/LinkedIn/GitHub

Research

Hardware/software co-design engineer specializing in large-scale HPC and AI systems, focused on the compute and communication bottlenecks that limit scaling of distributed AI workloads. Currently building FPGA/ASIC in-network accelerators and simulation infrastructure to characterize these bottlenecks in scale-up and scale-out MoE inference.

Education

- **University of Toronto** — Ontario, Canada
  Master of Applied Science - Computer Engineering — *Sep. 2025–Present*
  Advisor: Prof. Paul Chow
- **Universidad San Francisco de Quito** — Quito, Ecuador
  Bachelor of Science - Electrical Engineering — *Sep. 2020–Jul. 2025*

Experience

- **University of Toronto** — Ontario, Canada
  Summer Internships — *May 2024–Sep. 2024*
  - **High-Performance Reconfigurable Computing (HPRC)**:
    Ported a 100 Gbps FPGA-based Intrusion Detection and Prevention System (IDS/IPS) from Intel Stratix 10 to AMD Alveo U280 platforms.
  - **Space and Terrestrial Autonomous Robotic Systems (STARS)**:
    Designed a 3D Gaussian Splatting (3DGS) pipeline for non-fixed eye-to-hand calibration of robotic manipulators.
- **Synopsys Inc.** — Remote, Chile
  Bachelor's Co-op — *Nov. 2023–Nov. 2024*
  - **ASIC Physical Design**:
    Designed a 64-bit dual-mode logic (DML) ripple-carry adder with critical path delay error detection and correction (EDAC) in a TSMC 14nm FinFET process node.
- **National Polytechnic School of Ecuador** — Quito, Ecuador
  Research Assistant — *Jan. 2020–May 2024*
  - **Aeronautics and Thermofluids Applications (ATA)**:
    Developed a ROS-based simulation pipeline to test custom UAVs under extreme weather conditions for Andean wetland monitoring.

Publications

- **Conference**: Ona, G., Sanchez, A., Benitez, D. S. *Dragon AI: A Parameterizable AXI-Compatible FPGA Accelerator for Edge AI.* IEEE International Conference on Consumer Electronics (ICCE 2027). Under review.
- **Journal**: Valencia, E., Toapanta, F., Ona, G. et al. *An Open-source UAV Digital Twin Framework: A Case Study on Remote Sensing in the Andean Mountains.* Journal of Intelligent & Robotic Systems, 111, 71 (2025). doi:10.1007/s10846-025-02276-7

Projects

- **In-Network Acceleration for Distributed MoE Inference (Thesis Research)**
  - Built a discrete-event simulator to explore accelerator designs for distributed MoE inference. It sweeps model families across hardware configurations, network topologies, communication scheduling policies, and levels of parallelism to trace the dataflow at any element of the system.
  - Benchmarked Megatron-CUTLASS, Megatron-DeepEP, COMET, and FlashMoE implementations across 10 MoE models in intra- and inter-node tests on an H100 cluster with NVLink and NDR InfiniBand. Calibrated network and compute models to within 6% median error of measured dispatch, compute, and gather latencies.
  - Explored in-network acceleration with FPGA NICs offloading InfiniBand/PCIe traffic from GPUs and inline NVLink ASICs serving expert replicas under congestion. Averaged 1.9× (FPGA) and 4.3× (FPGA+ASIC) speedups over Megatron-CUTLASS at the first multi-node expert-parallel configuration.
- **FPGA-Based AI Accelerator with Reconfigurable Systolic Array Architecture**
  - Designed a weight-stationary systolic array for GEMMs from scratch, with a NoC-inspired row multicast controller that distributes weights. The host drives it through a custom 64-bit ISA over an AXI4-Lite interface.
  - Targeted consumer electronics with fully parameterized, reusable IP that scales from 4×4 to 256×256 arrays and adapts operand widths (INT8/16/32) to application needs.
  - Deployed and evaluated 11 array configurations on an AMD Kria KR260 FPGA, using 5–63% of LUTs and 14–174 mW of dynamic power.

- **Geo-Distributed Cloud Lab with FPGA Edge AI for Smart-Home Services**
  - Built a two-site (Ecuador–Canada) cluster over Tailscale with hierarchical DNS, Kubernetes orchestration, a TLS-secured service fabric, OpenBao key management, and an AWS EC2 proxy for global access.
  - Connected MinIO object storage replicated across sites, ClickHouse for telemetry analytics, Weaviate for vector storage, and a custom-PCB ESP32 sensor and lighting controller through a Kafka event bus.
  - Deployed people detection and face recognition on a Xilinx FPGA gateway running two Vitis AI DPUs in parallel on an RTSP camera stream. Faces are matched on the gateway and only events reach Kafka, so biometric data stays local.
- **VLSI CAD Flow: Technology Mapping, Partitioning, Placement, and Routing**
  - Built a C++ CAD backend for FPGA/CGRA architecture development, including ABC-based technology mapping, with a multithreaded visual frontend that renders progress live.
  - Developed a branch-and-bound partitioner and an analytical placer for heterogeneous DSP/BRAM cells. Evaluated FPGA architectures under different logic block and router connectivity.
- **Apache Spark vs. Flink Benchmarking for Distributed Data Processing**
  - Built an automated multi-node benchmarking framework with Kafka ingestion, NFS checkpointing, and Nexmark workloads to evaluate latency, throughput, scalability, and fault tolerance.
  - Demonstrated that Flink's record-at-a-time processing cut median latency by up to 45%, while Spark's micro-batches delivered up to 1.6× higher throughput. Spark also recovered from failures ∼7× faster by re-running only failed tasks, whereas Flink rolls the whole job back to a checkpoint.

Teaching

- **University of Toronto** — Ontario, Canada
  Teaching Assistant — *Jan. 2026–Present*
  - **ECE1756 Reconfigurable Computing and FPGA Architecture (Grad)**: Prof. Vaughn Betz
  - **ECE1779 Introduction to Cloud Computing (Grad)**: Prof. Chen Ying
  - **ECE352 Computer Organization (EngSci)**: Prof. Andreas Moshovos
  - **ECE243 Computer Organization**: Prof. Vaughn Betz & Prof. Jonathan Rose
  - **ECE342 Computer Hardware**: Prof. Karthik Ganesan & Prof. Parinaz Naseri
- **Universidad San Francisco de Quito** — Quito, Ecuador
  Teaching Assistant — *Sep. 2024–May 2025*
  - **IEE2003 Signals and Systems**: Prof. Luis Miguel Procel
  - **IEE3200 Semiconductors**: Prof. Luis Miguel Procel

Honors and Awards

- **Edward S. Rogers Sr. Fellowship** — Ontario, Canada
  University of Toronto – Graduate Fellowship — *2025–2027*
- **James Clerk Maxwell Scholarship** — Quito, Ecuador
  Universidad San Francisco de Quito – Undergraduate Scholarship — *2020–2025*
- **IEEE Distinguished Student Diploma** — Quito, Ecuador
  Universidad San Francisco de Quito – Electrical Engineering Department — *2023*
- **XXXVI Albert Einstein National Physics Contest** — Quito, Ecuador
  Bronze Medal — *2019*
- **XV Ecuadorian National Mathematics Olympiad** — Quito, Ecuador
  Bronze Medal — *2018*

Volunteer Experience

- **Student Volunteer, IEEE International Solid-State Circuits Conference (ISSCC)** — San Francisco, CA
  University of Toronto – Saratoga Group — *Feb. 2026*
- **Founder & President, IEEE Aerospace and Electronic Systems Society (AESS)** — Quito, Ecuador
  Universidad San Francisco de Quito – IEEE Student Branch Chapter — *2023–2025*

Skills Summary

- **Programming Languages**: C/C++, Python, Bash, CUDA, Verilog/SystemVerilog, Assembly (ARM/RISC-V/PTX).
- **Computer Architecture & Hardware Design**: Intel Quartus, Xilinx Vivado/Vitis/Vitis HLS/PetaLinux, Synopsys Design Compiler/ICC2, Cadence OrCAD (Capture/PCB Designer).
- **HPC & GPU Optimization**: NVIDIA Nsight (Systems/Compute), OpenMP, MPI, Slurm, PyTorch (DDP).
- **Networking & In-Network Systems**: NVLink, PCIe, Ethernet, AXI/Avalon, UART/I2C/SPI/USB, NCCL, FPGA networking, NoC, Tailscale.
- **Cloud & Distributed Systems**: Amazon Web Services, Docker (Compose), Kubernetes, Terraform, Ansible, Databricks, Apache Spark/Flink/Kafka/Zookeeper, Keycloak, OpenBao, NGINX, BIND9, Node.js.
- **Spoken Languages**: Spanish (Native), English (C1), French (B2).